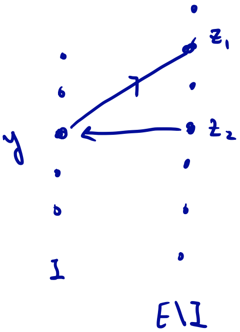
The Largest Common Ind Set

Given matroids $M_1 = (E, \mathcal{I}_1)$, $M_2 = (E, \mathcal{I}_2)$

find: $I \in \mathcal{I}_1 \cap \mathcal{I}_2$ maximizing $|I|$.

$D_{M_1, M_2}(I) = (E, A(I))$,



$I - y + z_1 \in \mathcal{I}_1$

$I - y + z_2 \in \mathcal{I}_2$

$X_1 = \{z \in E \setminus I : I + z \in \mathcal{I}_1\}$

$X_2 = \{z \in E \setminus I : I + z \in \mathcal{I}_2\}$

Case I: If $\exists$ $X_1 - X_2$ path in $A(I)$, then for P shortest path, $I' = I \triangle VP \in \mathcal{I}_1 \cap \mathcal{I}_2$.

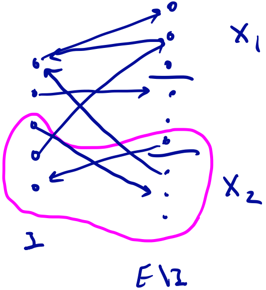
Augmenting Algorithm - Case II

Case II : $D_{M_1,M_2}(I)$ has no $X_1 - X_2$ path.

Thm : I is a max size common ind set.

pf :

Exercise : If no $X_1 - X_2$ path exists, then $\exists$ a set U with $U \cap X_1 = \phi$, $U \supseteq X_2$, and no arc enters U.



Proof — continued —

<u>Claim</u> : $r_{M_1}(U) + r_{M_2}(E \setminus U) \leq |I|$

(i) $r_{M_1}(U) \leq |I \cap U|$

If not, i.e., $r_{M_1}(U) > |I \cap U|$, then $\exists\, x \in U \setminus I$ with $I \cap U + x \in \mathcal{I}_1$.

However, $I + x \notin \mathcal{I}_1$ otherwise we would have $x \in X_1$, contradicting that $U \cap X_1 = \phi$

$I + x \notin \mathcal{I}_1 \Rightarrow \exists$ circuit $C \subseteq I + x$, with $x \in C$.

Moreover : $C \cap (I \setminus U) = \phi$ since $I \cap U + x \in \mathcal{I}_1$.

$\Rightarrow \exists\, y \in I \setminus U$ with $\boxed{I - y + x \in \mathcal{I}_1.}$

$\Rightarrow \exists$ arc that enters U. $\divideontimes$.

(ii) $r_{M_2}(E \setminus U) \leq |I \setminus U|$. Follows similarly.

This proves the claim, and optimality follows from Matroid Int Thm.